



LUT Business School

A220A6501 System Dynamics with Applications

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Project Plan: Data Brew Cafe

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1 Model

In order to assess the feasibility of opening up a small cafe in Helsinki, we have developed a System Dynamics Model that simulates an ongoing monthly financial projection based on a set of operational uncertainties and strategic decisions. The model is focused solely on one stock, the cash balance, and as such the cash balance is affected by the difference between the inflows and outflows each month, including the monthly inflows from the revenue generated by the cafe; monthly outflows for the operating costs of running the cafe; and the initial cash balance invested to start the cafe. The inflows (revenue) are dependent on the number of customers who visit the cafe each day, the amount spent by each customer and the number of days the cafe operates in each month (24 days, Monday to Saturday). The outflows (operating costs) include the fixed rental cost of the location of the cafe, the raw materials and other ongoing operating costs of the cafe.

These relationships were defined with a stock and flow model structure. The uncertainty was defined through probability distributions and sampling methodologies, as well as through Global Sensitivity Analysis. This will provide a methodical process to determine the impact of demand, price and costs on the cash flows of the cafe over time. In addition to the Global Sensitivity Analysis, we used SimDec Decomposition to verify the consistency of our assumptions and to identify the most important variables, as well as to determine the conditions under which the cafe remains financially viable. The model contains both uncertainty and decision variables. Uncertainty variables include the following. Daily Customer traffic the model assumes 35 to 70 customers per day. It can be said that Business Dojo (2025) backs this range, by mentioning that a normal new café in Finland could expect 35 to 60 customers per day, with some cafés reaching even more, like 100 or more during peak hours. Average spending per customer can vary as well. Average spending per customer also varies. The model uses a range of 6 to 10 euros, in line with prices for coffee and snacks in Helsinki reported by Global Price Info (2024). Raw material costs vary between 25 percent and 35 percent of total sales, typical for the food service industry (Barista Life, 2024). The monthly rent could range values from 900 to 1200 euros, as backed by KVL Finnish Property Data (2024). The initial investment, also important to mention, could range from 28,000 to 38,000 euros, which seems to be consistent with the estimates for a lean setup (Restroworks, 2025).

2 Model Analysis

The café's financial position is largely defined by daily customer numbers and average spend per customer - these have the greatest impact on the 24-month financial position and represent the largest positive and negative impacts shown in the tornado plot (each of the two impacting the 24-month balance by approximately +/- 35k euro for a 20% change). This is also evident in the heat map; this shows an increasing, flat-surfaced relationship between customer traffic and profit, such that a sharp

increase in profit occurs when the number of customers exceeds 60/day and the spend increases above 9-10 euro. The results from the scenario analysis confirm these relationships; an optimistic scenario of 70 customers/day and 10 euro spend produces a 24-month balance of 252.9k euro compared to the 35 customers/day and 6 euro spend pessimistic scenario of 11.8k euro. Thus, the potential outcomes of the café appear to be very sensitive to both demand and pricing. A Monte-Carlo simulation that considers all the uncertain elements simultaneously provides a centre-distribution of mean $\approx$ 114k euro with 0% probability of a negative outcome at the end of 24 months. Although the individual effects of Cost of Goods Sold, Rent and Investment uncertainty are important, the overall effect of these elements is relatively small compared to the major influences of customer volumes and expenditure patterns. In summary, the four analytical methods used provide evidence of a consistent relationship: that the inter-relation of the revenue generating variables will reinforce one another and provide most of the variability of outcomes and that the variability of costs will determine the range of possible outcomes but not the inherent viability of the business.

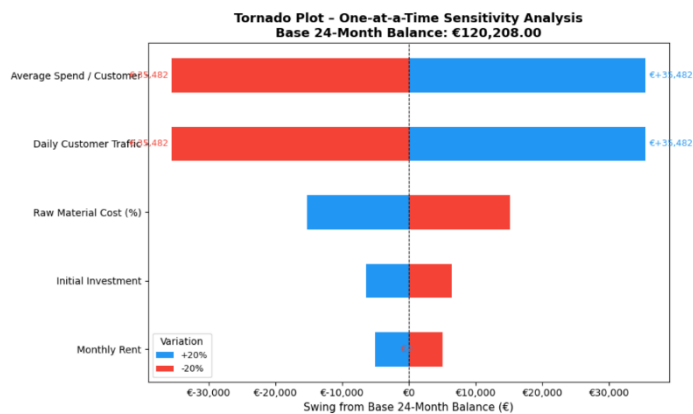


Image 1: Tornado Plot

Sensitivity Results:

variable	direction	value	balance	swing
customer_traffic	high	66.00	155689.6	35481.6
customer_traffic	low	44.00	84726.4	-35481.6
average_spending	high	9.60	155689.6	35481.6
average_spending	low	6.40	84726.4	-35481.6
raw_material_cost	high	0.36	105001.6	-15206.4
raw_material_cost	low	0.24	135414.4	15206.4
rent	high	1260.00	115168.0	-5040.0
rent	low	840.00	125248.0	5040.0
initial_investment	high	38400.00	113808.0	-6400.0
initial_investment	low	25600.00	126608.0	6400.0

Image 2: Sensitivity Results(Tornado Plot.)

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SCENARIO ANALYSIS - 24-MONTH BALANCE

Fixed: Location=Neighborhood | Staff=1 | Model=Cafeteria only

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— Input Variables —

Scenario	Daily Traffic	Avg Spend (€)	Raw Material Cost	Monthly Rent (€)	Initial Investment (€)
Optimistic	70 customers	€10	25%	€900	€28,000
Expected	55 customers	€8	30%	€1,050	€32,000
Pessimistic	35 customers	€6	35%	€1,200	€38,000

— Output —

Scenario	Location	Staff	Business Model	24-Month Balance (€)
Optimistic	Neighborhood	1	Cafeteria only	€252,800.00
Expected	Neighborhood	1	Cafeteria only	€120,208.00
Pessimistic	Neighborhood	1	Cafeteria only	€11,824.00

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Image 3: Scenario Analysis

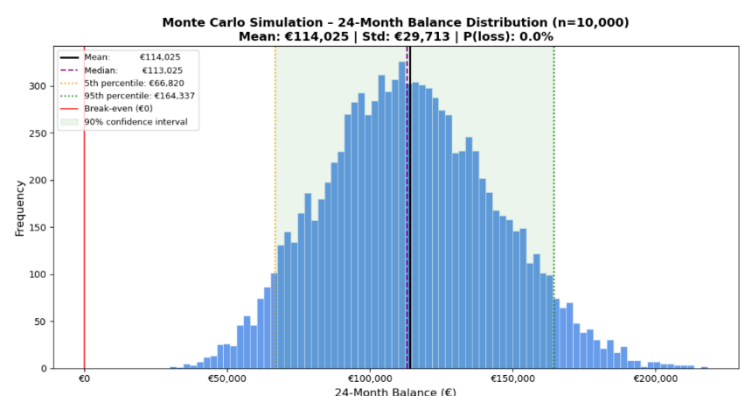


Image 4: Monte Carlo Simulation

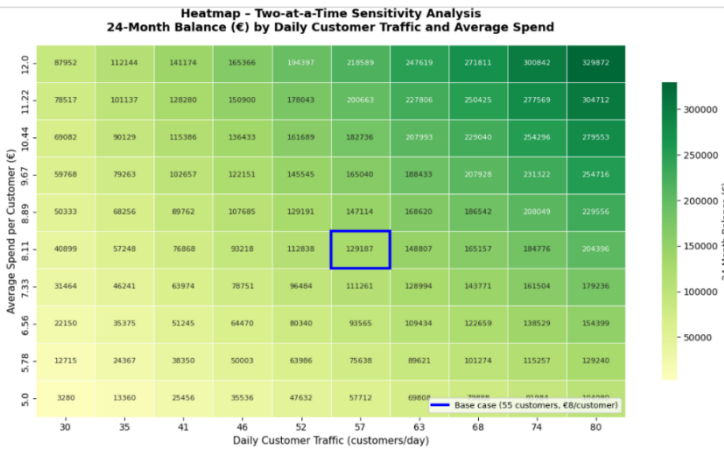
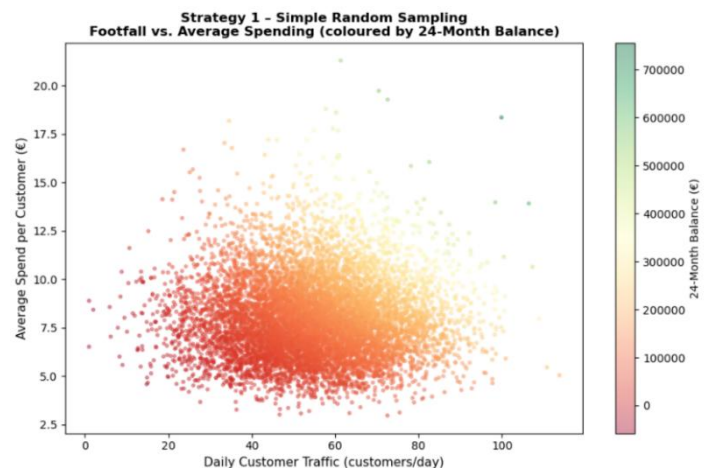


Image 5: Heat Map (Average Spend Per customer vs Daily customer Traffic)

3 Modelling Uncertainty and Sampling

The probability distribution of customer traffic was modelled with a truncated normal distribution having a mean of 55 and a standard deviation of 15, which is consistent with customer volume of new cafes in Finland (Business Dojo, 2025). The expected average spend on each customer followed a log-normal distribution based on an average of 8 euro, with a standard deviation of 2 euro; which is typical of the coffee and snacks sold in Helsinki (Global Price Info, 2024); (Cost to Travel, 2025). The cost of raw materials expressed as a % of sales was modeled with a Beta distribution with min = 25%, max = 35% and mod = 30%; this range is representative of the cost of goods sold for the food service industry (Barista Life, 2024). Rent costs were modeled as a uniform distribution between €900 and €1300; which represents a reasonable commercial property price range in Helsinki (KVL Finnish Property Data, 2024). All four uncertainty distributions create a realistic space of possible inputs into the simulations. A simple random sampling method produced a densely populated operational area around 50 to 70 customers and €6 to €10 average spend; increasing profitability at both points as they increased. Sobol Quasi-Monte Carlo sampling increased the number of covered spaces by decreasing the clustering of the samples and enabling the capture of a greater number of extreme combinations of the two variables. The partial factorial plus random sampling method demonstrated that while all three decision variables (location, staff, menu) have similar demand areas, the financial results differ significantly depending upon the cost structure used for each variable.

Image 6: footfall vs. average spending



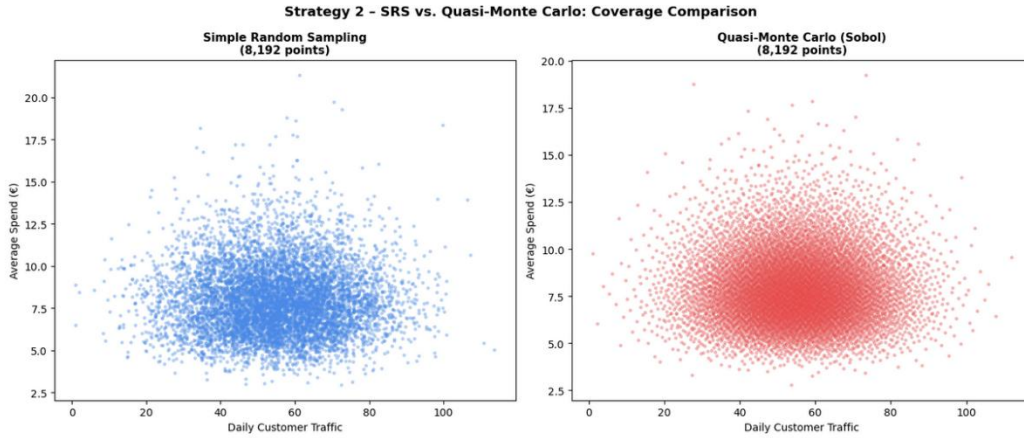


Image 7: Scatter plot of footfall vs. average spending, compare with simple random sampling

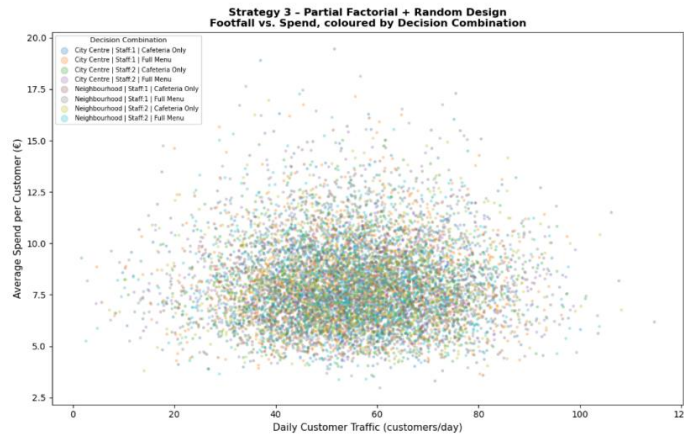


Image 8: Inflow vs. average expenditure

4 Global Sensitivity Analysis

Customer traffic and average spend are the two most influential variables in the global sensitivity analysis for variability in the 24-month financial balance. Customer traffic explains 49.5% of total output variance, and average spend explains 40.8%, while all other variables explain less than 1%. Therefore, regional sensitivity analysis results are also very similar to those of the global sensitivity analysis, where both customer traffic ($KS = .9472$) and spending ($KS = .3772$) are the only variables that show meaningful separation between successful and unsuccessful outcomes. Therefore, initial investment or rent can be seen as mild discriminators at the break-even point, however they do not have any material impact on overall variance. Therefore, both methods clearly support the concept that the behaviour of this system is predominantly driven by demand-side dynamics, which supports earlier analyses and clearly indicates that pricing strategy and customer acquisition efforts should be the top priority.

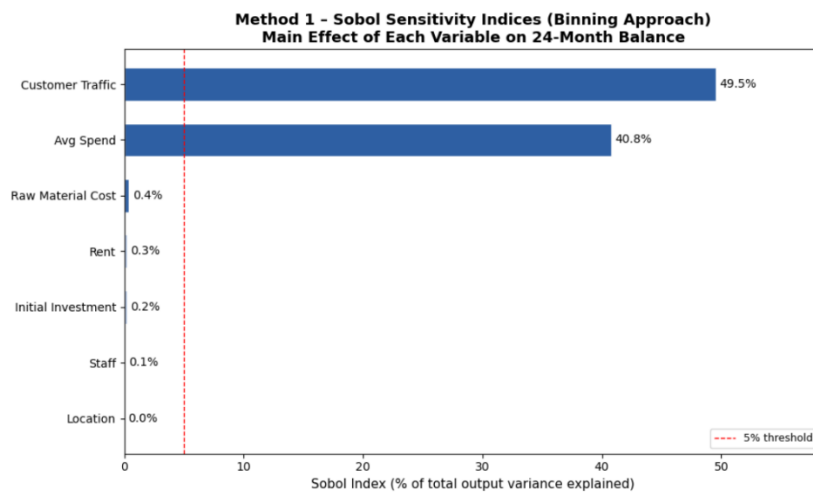
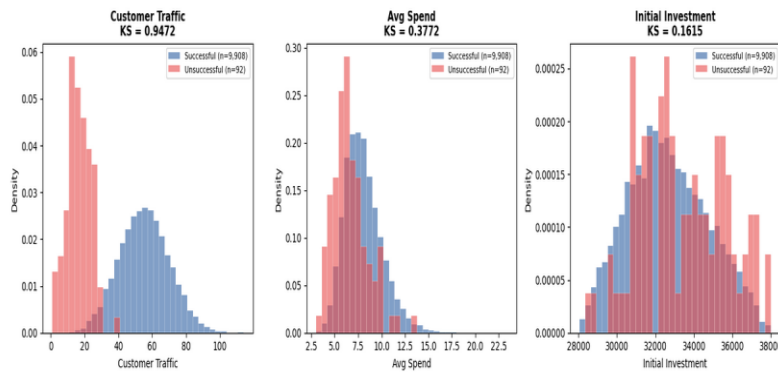


Image 9: Sobol Sensitivity Indices

Method 2 - Regional Sensitivity Analysis
Top 3 Discriminating Variables: Successful vs. Unsuccessful Outcomes



METHOD COMPARISON - SOBOL vs RSA RANKINGS

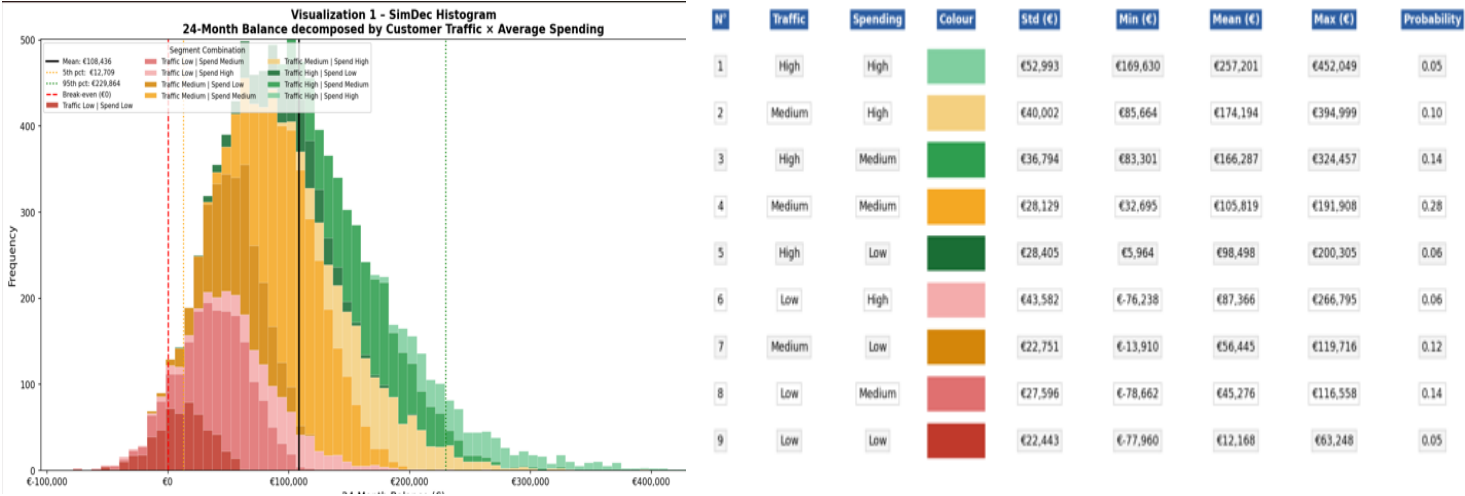
Variable	Sobol Rank	Sobol Index (%)	RSA Rank	KS Statistic
Customer Traffic	1	49.53	1	0.9472
Avg Spend	2	40.80	2	0.3772
Raw Material Cost	3	0.41	5	0.0436
Rent	4	0.26	4	0.1099
Initial Investment	5	0.25	3	0.1615
Staff	6	0.06	6	0.0212
Location	7	0.04	7	0.0061

Image 10 &11: Regional Sensitivity Analysis (Monte Carlo filtering) & Regional Sensitivity Analysis (Monte Carlo filtering)

5 Visualization

The histogram for the SimDec package (Kozlova and Yeomans, 2024) in python shows that the cafe's 24 months financial performances are extremely dependent upon the combination of traffic and spending of customers. As the variables of traffic and spending move from low to high, the histogram shifts to the right. The best performing area is where there is a high volume of customers with high spending; this is because it produces the largest average balance and always produce positive financial outcomes. Both Medium Traffic $\times$ High Spending and High Traffic $\times$ Medium Spending have strong financial outcomes and have little downside. However, the three low traffic areas are clustered around and below zero and have the lowest possible positive tail, showing that too few customers will drive risk financially. The two simulation output scatter plots support this conclusion. Nearly all locations improved their year over year financial outcomes, however, the level of staffing and location influence the financial outcome only when enough customer traffic has been created. These conclusions clearly demonstrate that demand conditions determine the viability of an operation. Therefore, the desired combinations are those that provide at least medium volumes of customers with medium to high

spending. All other low-traffic combinations present structural financial risk. (Kozlova and Yeomans, 2024).



Images 12 & 13: 24-Month Balance decomposed by Customer Traffic & Average Spending & Probability Distribution of the combinations.

6 Conclusion

Across all analyses, the Café model has a consistent approach to the business. The business will be financially viable for many realistic scenarios (if both customer traffic and the average spend by those customers stay moderately high) and very profitable if both are high. The deterministic Scenario Analysis and Tornado Plot show that the variables related to demand have the most effect on how well the business performs. The same conclusion is shown in Uncertainty Modeling and Global Sensitivity Analysis. Traffic and spending, which are two of the demand variables, explain more than 90% of the variability in the outputs and are the only two variables that consistently separate successful from unsuccessful outcomes. The sampling-based exploration, including Sobol Sequences and Factorial Designs, illustrate that the structure of the business (e.g., number of employees or location) does not significantly affect the outcome until demand has been established. Although there is some variation in the long-run balance trajectories due to cost-side uncertainty, the effects are generally minor. The SimDec Decomposition also shows the same pattern. High or medium traffic with medium to high spending moves the distribution of outcomes toward more positive and stable. The low-traffic regimes tend to cluster around or below zero indicating insufficient footfall as the major structural risk. In summary, sustaining financial viability will depend on establishing an ongoing and reliable stream of customers and ensuring each customer spends enough money. Decisions about operational and costs will improve the level of performance; however, they will not define the direction of the overall path. As founders, we believe these results provide us the confidence to move forward in pursuing our dream of launching Data Brew and providing sustainable revenue streams.

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Appendix

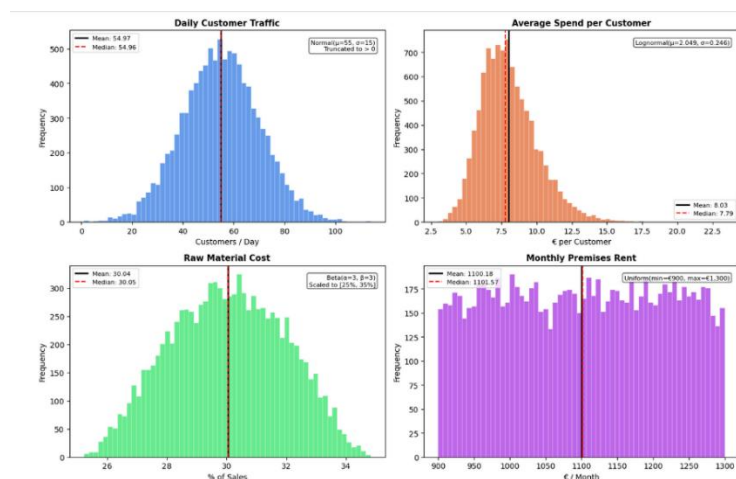


Image 14: Respective Distribution types (Uncertainty & sampling)

DISTRIBUTION SUMMARY STATISTICS (n=10,000)	
Customer Traffic	
Mean:	54.969
Std:	15.048
Min:	1.000
Max:	113.894
Average Spending	
Mean:	8.027
Std:	2.000
Min:	3.003
Max:	23.382
Raw Material Cost	
Mean:	30.043
Std:	1.900
Min:	25.223
Max:	34.824
Monthly Rent	
Mean:	1100.179
Std:	114.691
Min:	900.052
Max:	1299.947

Image 15: Distribution Summary Statistic (Uncertainty & sampling)

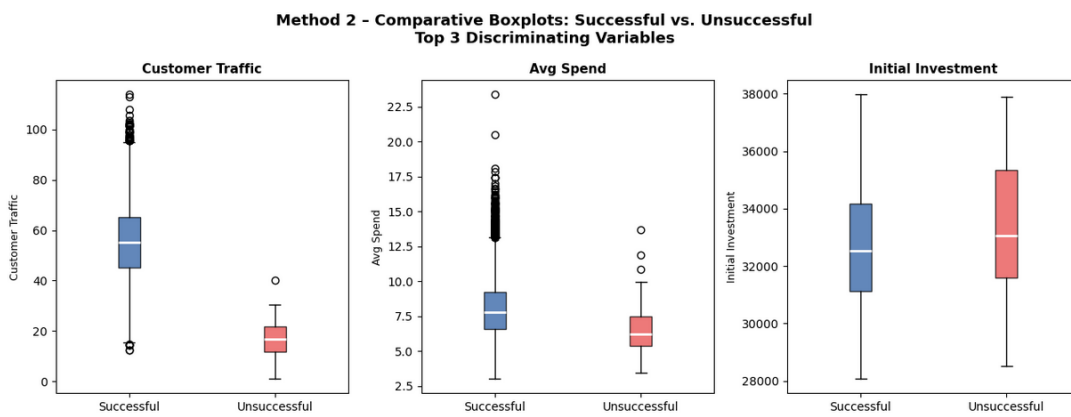


Image 16: Regional Sensitivity Analysis (Monte Carlo filtering)

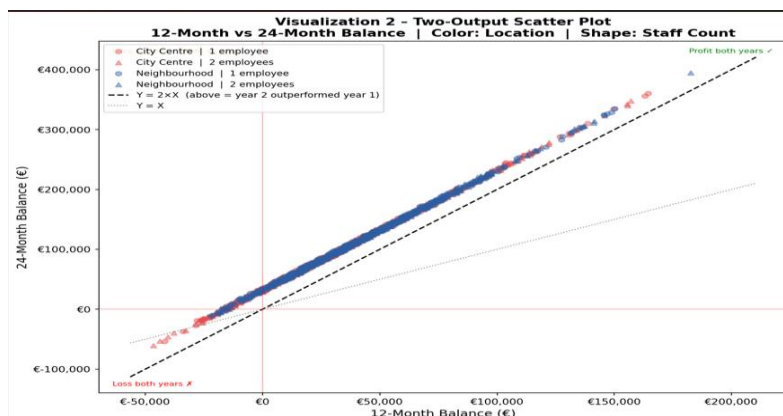


Image 17: 12-Month vs 24-Month Balance.